FERN Network Parameters

February 2025

1 Intro to the thermonuclear networks

There are several networks that are able to be run using FERN, not all are supernova related. Most of these networks are simulating reactions that occur at various points in the star's lifetime. I will try to list them in order from early/main sequence lifetime reactions to reactions that occur during late stages of evolution or even the death of a star.

2 Main sequence reactions

These are the reactions that occur during the main sequence life of a star, they convert hydrogen (H) to helium (He). The first one is the proton proton chain. This occurs in stars with less than about 1.3-1.5 M$\odot$. Once the cores of stars reach temperatures of about 4 million K, the proton-proton chain is able to fuse 4 hydrogen nuclei (or just 4 protons) to a He nucleus. This is the dominant source of energy up until about 16-20 million K, when the CNO takes over.

The second network is the CNO network. As mentioned above, the CNO reactions become the dominant source of energy production in stars when the temperatures reach 16-20 million K. Because higher temperature are needed for the CNO to be efficient enough to be the dominant energy source, it is present in stars with mass $M_* > 1.3 - 1.5M_\odot$. The more massive stars are able to reach higher temperatures in their cores, so that allows the CNO cycle to kick in.

2.1 P-P chain

- ISOTOPES: 7
- SIZE: 28
- Network and Reactions files: pp.inp and pp.data
- Hydro Profile: Not presently used
- Methods: ASY and ASY+PE (QSS once that works as well as QSS+PE then too)

- T9 = 0.01 - 0.02, realistically the p-p chain occurs once core temps reach about 4 million (T9 = 0.004), but the code hangs up and does not appear to evolve the network if T9 < 0.01
- ρ = 75 to 1000 g/cm^3, typical values is 1e2, but the more massive stars can have the higher densities
- Tolerances: For fast cases $1e-3 \rightarrow 1e-4$, Intermediate cases $1e-5 \rightarrow 1e-7$, Accurate $1e-8 \rightarrow 1e-10$ For now keep the masstolASY and masstolPE set to be the same values.
- Start time: 1e-20;
- Stop time: $1e19 \rightarrow 1e22$
- Start Plot time: $1e6 \rightarrow 1e10$

2.1.1 Astrophysical Relevance

The p-p chain is the dominant source for low mass stars (typically those less than about 1.3-1.5 M$\odot$). There are four branches of it, but the main one is shown below.

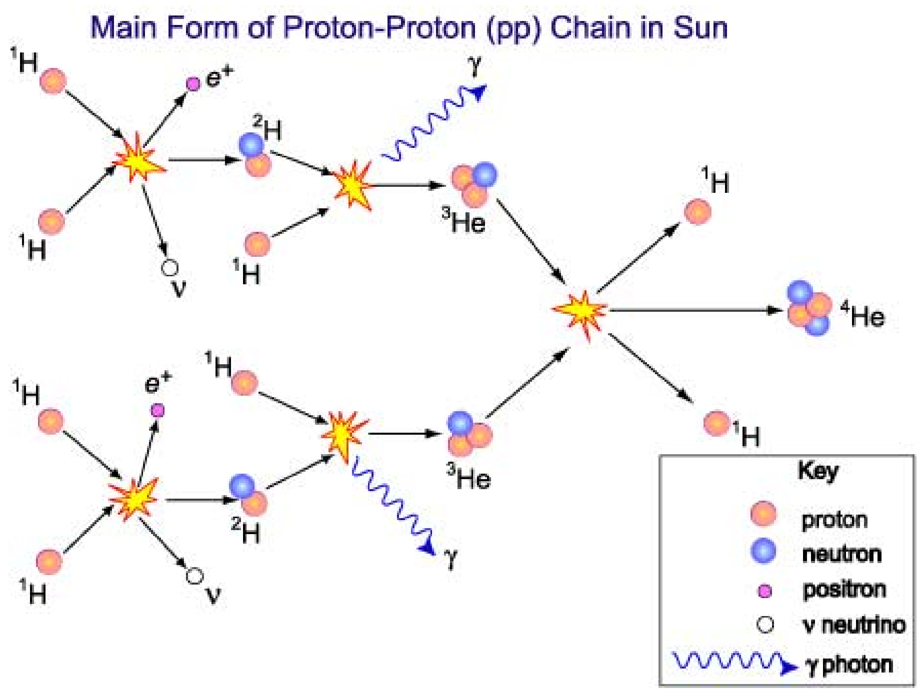

Figure 1: Main branch of P-P chain

2.2 CNO Network

Another type of network that simulates the main-sequence life of a star slightly more massive than our sun. On the slower end the CNO cycle will run into H-depletion on the order of 1-10 Billion yrs ($10^{17} - 10^{18} s$). The CNO network is heavily dependent on temperature ($Rate \propto T^{17}$), so the time it reaches H-depletion will depend on your initial value for T9.

There are 2 networks that run the CNO cycle: Main CNO (core isotopes) and full CNO (all 16 isotopes)

- ISOTOPES: 16 (full) or 8 (main)
- SIZE: 134 (full) or 22 (main)
- Network and Reactions files: CNOall.inp and CNOall.data (Full) or cno.inp and cno.data (Main)
- Hydro Profile: Not presently used
- Methods: ASY and ASY+PE to a lesser extent (until QSS is working)
- T9 = 0.015 - 0.03, our typical value is 0.02, but the CNO starts to dominate around 15 million K
- ρ = 100 to 1000 g/cm^3, though more massive stars can go up more
- Tolerances: For fast cases $1e-3 \rightarrow 1e-4$, Intermediate cases $1e-5 \rightarrow 1e-7$, Accurate $1e-8 \rightarrow 1e-10$ For now keep the masstolASY and masstolPE set to be the same values.
- Start time: 1e-20;
- Stop time: $1e16 \rightarrow 1e18$
- Start Plot time: $1e5 \rightarrow 1e7$

2.3 Astrophysical Relevance

Another way to convert 4 H nuclei (proton) to He nuclei (alpha particle). Typically not the dominant source of energy until stars get to about $1.5M_{\odot}$, otherwise the proton-proton chain is the dominant source of energy (different network).

The CNO cycle is heavily temperature dependent ($rate \propto T^{17}$), so higher temps will cause H-depletion much quicker than lower temps. Around 15 million K is when the CNO starts becoming more efficient. The rate if fusion increases faster in these stars which is why massive stars are much brighter and hotter than low-mass stars. They also brun through their H source much quicker and have shorter main sequence lives. To see this, tr running the CNO network with only changing the temperature. When the temp is around T9 = 0.015 or 0.02, the H abundances do not dropoff until about 10^{16}s or so. If T9 = 0.03, the H abundances drop around 10^{13}s.

Steps of the CNO:

1. A carbon-12 nucleus captures a proton and emits a gamma ray, producing nitrogen-13.
2. Nitrogen-13 undergoes β-decay into carbon-13.
3. Carbon-13 captures a proton and becomes nitrogen-14 via emission of a gamma-ray.
4. Nitrogen-14 captures another proton and becomes oxygen-15 by emitting a gamma-ray.
5. Oxygen-15 undergoes β-decay into nitrogen-15.
6. Nitrogen-15 captures a proton and undergoes alpha decay to produce a helium nucleus (alpha particle) and carbon-12, which is where the cycle started.

In this process, energy is produced in the form of gamma rays (primarily), which eventually radiate outward from the star. The radiation pressure outwards provides the energy that counteracts the inward force of gravity and supports the star against collapse.

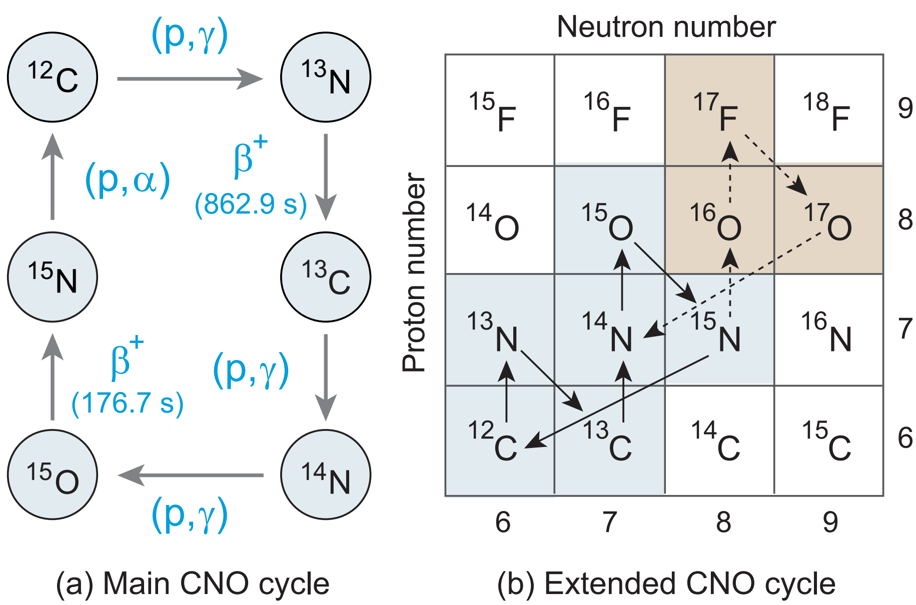

Figure 2: Schematic of CNO cycle with participating isotopes

3 Late stages

3.1 Triple Alpha Network

This is the simplest network we run. Generally recommended for debugging or for testing new code that we are looking to implement on larger networks. This would be a good network to use to fix the issue with QSS since it is simple and the errors can be tracked down more easily.

It has 3 isotopes He4, C12, and O16. This reaction process forms the Carbon and Oxygen in stars from He4+He4 $\rightarrow$ Be8. Be8 is unstable and decays within $8 \cdot 10^{-17}$s, so another He4 has to fuse with it almost simultaneously. An additional He4 can fuse with C12 to create O16. This reaction is strongly dependent on temperature and not as much on density, $Rate \propto T^{40}$ and ρ^2.

- ISOTOPES: 3 or 4
- SIZE: 8 or 14
- Network and Reactions files: 3alpha.inp and 3alpha.data OR 4alpha.inp and 4alpha.data
- Hydro Profile: Typically no
- Methods; ASY and ASY+PE (until QSS is working)
- T9 = 2-9, we typically use 5-7 though
- ρ = 1e6 to 1e8 g/cm^3, though we typically run it at 1e8
- Tolerances: For fast cases $1e-2 \rightarrow 1e-4$, Intermediate cases $1e-5 \rightarrow 1e-6$, Accurate $1e-7 \rightarrow 1e-10$

  For now keep the masstolASY and masstolPE set to be the same values.
- Start time: 1e-20;
- Stop time: $1e-4 \rightarrow 1e-2$
- Start Plot time: $1e-16 \rightarrow 1e-14$

3.1.1 Astrophysical Relevance

Since this reaction lacks H, it is generally present later on in stellar evolution during He burning. For stars with at least 0.8 solar masses, once the core temperature becomes high enough (around 100 million K or T9 = 0.1), the triple-alpha process becomes the dominant way of generating energy and producing heavier elements.

The production of carbon via through the triple-alpha process can have a significant impact on the star's structure and evolution. The presence of a C/O core with a shell of He burning influences pulsations, mass loss, and instabilities in the later stages of stellar evolution.

In a more massive star (like a red giant), the core becomes predominantly carbon and oxygen after the triple-alpha process, while the outer layers of the star go through hydrogen and helium shell burning. The C-O core is what most low mass stars end as, since they can't fuse C or O into heavier elements. During the He burning phase, the star has blown off its outer layers, so the C-O core is the only remnant and it is a white dwarf "star". Since fusion stops, there is no longer energy being produced in the core and it becomes inert, supported against collapse by electron degeneracy pressure.

This is the source of the type Ia supernova under the single degenerate model. A COWD (Carbon-Oxygen White Dwarf) is supported against collapse by electron degeneracy pressure up to about $1.44M_{\odot}$ (Chandrasekhar Limit). If a companion star (usually a red giant), fills its Roche Lobe and begins to accrete matter on to the WD, the WD will begin to increase in mass. Once enough matter has accreted to raise the pressure and temperature in the core, carbon ignition occurs. This event is very energetic and happens quickly. The rapid fusion of Carbon leads to an explosive run away reaction which destroys the WD and gives us the Type Ia supernova.

3.2 Alpha Network

This network has typically been the one most people use, especially in hydro codes. It's a simple approximation using the more important isotopes. Network sizes are typically limited for computational reasons, so the alpha network has been a simple network to use in hydrodynamic codes. Use this as a baseline test once new code has been tested under the 3-alpha, typically if there is something that can go wrong it will go wrong in this network.

The alpha network consists of isotopes that are a multiplicity of He4 (C12, O16, Ne20, Mg24, Si28, S32, Ar36, Ca40, Ti44, Cr48, Fe52, Ni56, Zn60 and Ge64).

- ISOTOPES: 16
- SIZE: 48
- Network and Reactions files: alpha.inp and alpha.data
- Hydro Profile: Use $ViktorProfile_{4}00.inp$ if wanting to use one. – **max hydro entries = 408**
- Methods; ASY and ASY+PE (until QSS is working)
- T9 = 2-9, we typically use 5-7 though
- ρ = 1e6 to 1e8 g/cm^3, though we typically run it at 1e8
- Tolerances: For fast cases $1e-3 \rightarrow 1e-4$, Intermediate cases $1e-5 \rightarrow 1e-7$, Accurate $1e-8 \rightarrow 1e-10$ For now keep the masstolASY and masstolPE set to be the same values.
- Start time: 1e-20;
- Stop time: $1e-3 \rightarrow 1e0$
- Start Plot time: $1e-16 \rightarrow 1e-14$

3.2.1 Astrophysical Relevance

This reaction network is typically occurs in stars that are in later stages of evolution, typically during stellar explosions (such as supernovae) or in very high-temperature environments like the interiors of massive stars.

The basic idea is that alpha particles (He nuclei) fuse with other other isotopes (multiples of He4) to form new, heavier elements. The process primarily occurs in the core of massive stars, where extremely high temperatures and pressures allow the fusion of heavier nuclei.

After a star depletes the H in its core, it enters the He-burning phase. During this time He fuses into carbon and oxygen via the triple-alpha process. After C/O it then goes to Ne and Mg and up the alpha ladder above.

For stars with masses greater than 8-10 times the mass of the Sun, the core temperature becomes high enough to allow alpha capture reactions, which leads to the formation of heavier elements like silicon, sulfur, and eventually iron. The star becomes a supergiant, and the core undergoes a series of shell fusion stages where each shell is responsible for fusing progressively heavier elements. In these stars, the core will have layers where different elements are being fused at different temperatures (O-shell, C-shell, He-shell etc...). The outer shells typically fuse lighter elements (H and He), while the inner layers are responsible for the fusion of heavier elements through the alpha process. The "onion model" where each layer of star is a shell that is fusing successively heavier elements (moving from outer layers to inner layers). Note: This is just a conceptual model, not meant to be taken as reality.

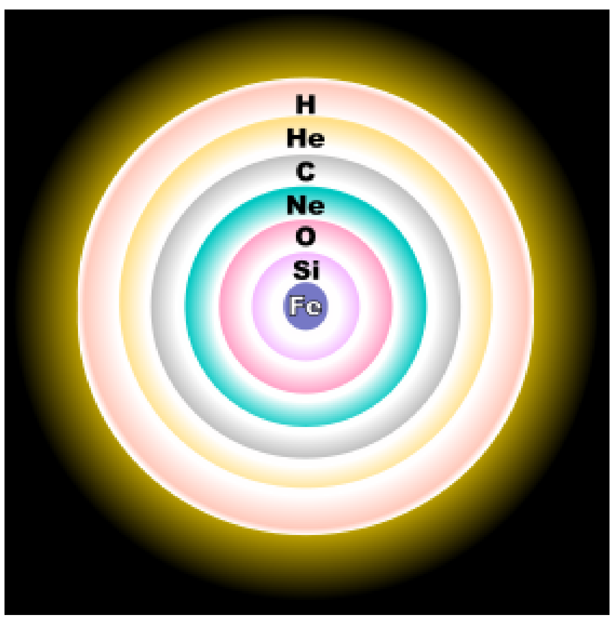

Figure 3: Onion Shell model of Massive star at end stages of life with Fe in the core fusing progressively lighter elements moving out.

In the violent explosion of a supernova, the temperature and pressure become so high that alpha capture continues in the outer layers of the star. This process helps form a wide range of elements, including those up to Fe and Ni. More relevant for CCSN.

3.3 Nova Network

A more realistic network that populates 134 isotopes and involves 1500+ reactions. We use a hydro profile to vary T and ρ and don't typically use constant T and ρ but that could be explored some. This is a more realistic representation of the isotopes that are populated in typical nova explosions. The explosion occurs around $10^{1-2}s$, where all the isotopes are participating in reactions. The figures below help bring a complete picture together. The first one is the mass fraction vs time, where the explosion occurs where all the chaos is. The hydro profile shows the temperature and density over time. At the same time the explosion occurs, the temperature spikes (the reason for the explosion), and the density drops (since material is being ejected outward).

- ISOTOPES: 134
- SIZE: 1566
- Network and Reactions files: nova.inp and nova.data
- Hydro Profile: NovaDProfile125.inp – **max hydro entries = 403**
- Methods: ASY and ASY+PE to a lesser extent (until QSS is working)
- T9 = We don't use constant T for this
- ρ = Also don't use constant density either
- Tolerances: For fast cases $1e-3 \rightarrow 1e-5$, Intermediate cases $1e-6 \rightarrow 1e-7$, Accurate $1e-8 \rightarrow 1e-10$ For now keep the masstolASY and masstolPE set to be the same values.
- Start time: 1e-20;
- Stop time: $1e6 \rightarrow 1e8$
- Start Plot time: $1e-5 \rightarrow 1e-7$

3.3.1 Astrophysical Relevance

If a star has a mass of 7-8 $M_\odot$, it will end it's life as a white dwarf. The composition of the WD depends on the initial mass (or ZAMS mass - zero age main sequence mass). The least massive stars fuse H to He and then cannot meet requirements to fuse He to heavier elements. These stars end their lives as He white dwarfs (HEWD), though theoretically, none should exist yet as their main sequence life is longer than the current age of the universe. The more massive stars are able to fuse He in their cores and reach C and O. If the conditions in the core are not sufficient to fuse C and

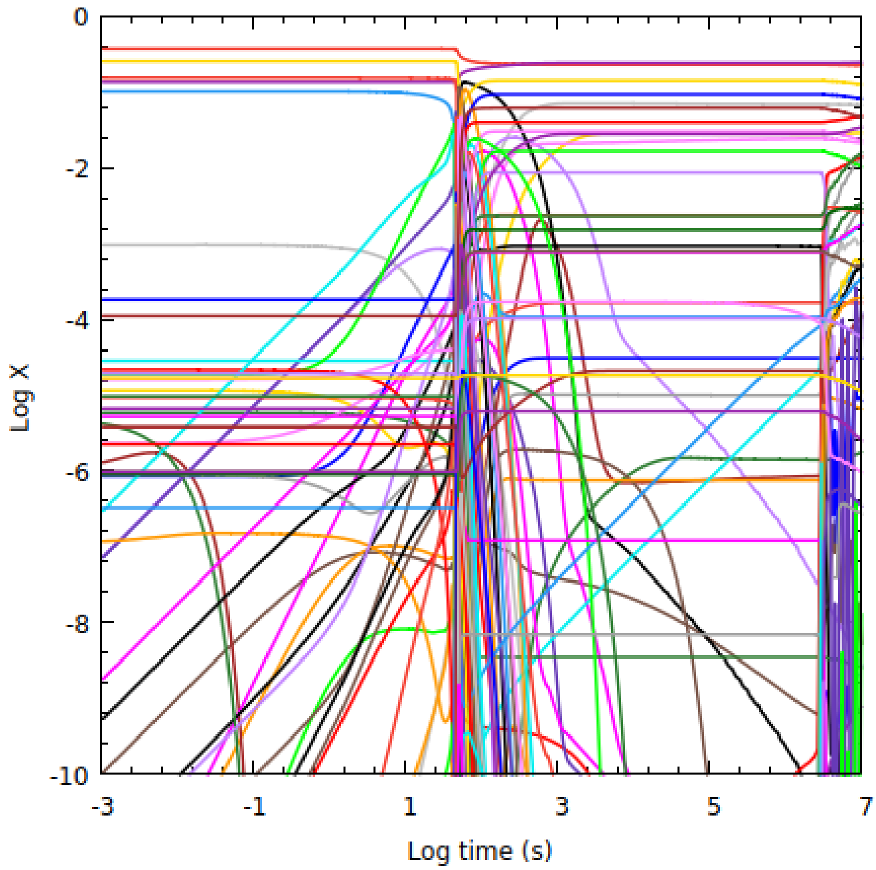

Figure 4: Plots of the Mass fraction evolution over time

O into heavier elements, these stars will end their lives as C-O white dwarfs (COWD). The most massive stars that do not undergo core collapse (pushing the limit close to the 8M$\odot$ now) are able to fuse C and O in their core and can produce O,Ne and possible Mg. Any more massive stars tend to be able to fuse up to Fe and then undergo core collapse to form neutron stars or black holes. This network is more so for a COWD, though, so a star similar to our sun.

Once a star is unable to have fusion in its core, there is no longer an outward radiation pressure to balance the inward force of gravity. The star contracts until electron degeneracy provides the pressure to support against further gravitational collapse. If the star is in a single star system, it'll radiate it's remaining heat away but it will not be able to further generate heat and it slowly cools off. If it part of a binary system, then it opens the possibility of a nova or supernova.

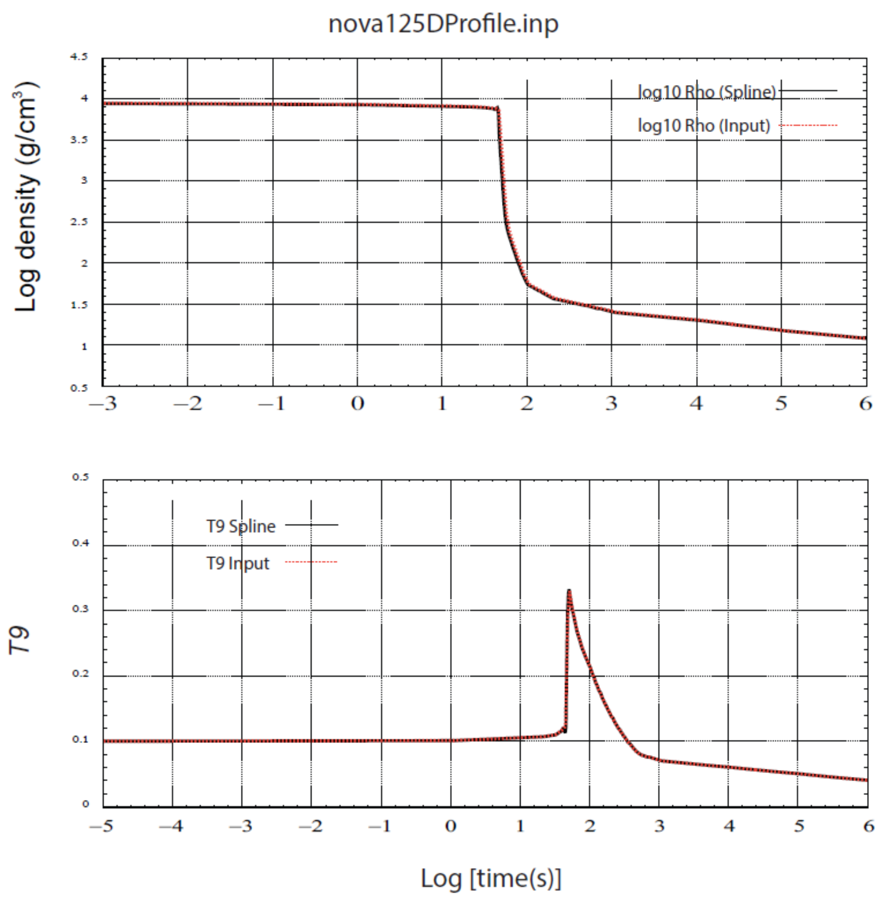

Figure 5: Temperature and Density evolution over the course of the calculation

3.4 Nova explosion

As the WD's companion star evolves and reaches it's red giant phase, matter is able to be transferred from the companion over to the WD. The matter (mostly H and some He) will begin to accrete on the surface of the WD. Once the accreted hydrogen becomes hot (around $10^6 K$ and dense enough, it will ignite and thermonuclear fusion of H occurs on the surface of the WD. This releases a large amount of energy is a short time and the result is a nova explosion (which is an outburst of light and material being ejected from the surface of the white dwarf). In a nova explosion, the WD remains intact as the explosion was due to ignition of the H on it's surface.

3.5 Type Ia Supernova

There are 2 models for type Ia supernovae, the first is related to the nova, where a WD accretes enough matter to ignite the core and the WD explodes (single degenerate model). The other is the WD's companion is also a WD and the two WDs merge and create a type Ia supernova (double degenerate model). In the single degenerate model, the WD accretes material (H) from it's companion red giant star. The electron degeneracy pressure prevents the star from further gravitational collapse, but only if the WD's mass is less than the Chandrasekhar limit (about 1.45 $M\odot$). If enough material is accreted from the companion star to increase the mass of the WD above the limit, the electron degeneracy pressure can no longer balance out the inward force of gravity. The core is degenerate, so increasing temperature does not cause the star to expand, and density remains roughly the same. At a certain point, the C and O in the core reach the necessary temperature to ignite fusion. Runaway nuclear fusion occurs throughout the entire body which leads to an explosive event, causing the white dwarf to completely destroy itself in a supernova explosion.